COURIER MANAGEMENT - DOCUMENTATION PHASE

Project Overview (Page 1 of 3)

Project Abstract

Problem: Manual tracking difficult
Solution: Online Courier System
Users: Customer, Agent, Admin
Features: Booking, Tracking, POD

Scope & Objective

Online Parcel Booking
Real-time Tracking CMS123456
Agent Assignment Auto
Invoice & POD Generation

System Requirements

Hardware: 4GB RAM, i3
Software: Python, MySQL
Browser: Chrome, Firefox
Tools: XAMPP, VS Code

Methodology

Agile - 1 Week Sprint
4 Members Division
Daily Standup Meeting
GitHub Version Control

ER & Database

5 Tables: Users, Shipments
Tracking_History, Agents
Invoices, POD Tables
3NF Normalized Design

Architecture

3-Tier: UI, API, DB
8 REST API Endpoints
Bootstrap Responsive
Secure Auth with Hash

COURIER MANAGEMENT - MODULES DOCUMENTATION

6 Modules Detailed (Page 2 of 3)

Module 1: User Mgmt

Registration & Login
Role: Customer/Agent/Admin
Password Hash bcrypt
OTP Verification

Module 2: Booking

Sender/Receiver Form
Weight & Distance Input
Cost Calc: weight*rate
Tracking ID CMS+6digit

Module 3: Tracking

Input Tracking ID
Timeline UI
Location Update
History Table Query

Module 4: Agent

View Assigned Orders
Update Status Button
POD: Photo Upload
OTP from Customer

Module 5: Admin

Dashboard Analytics
Assign Nearest Agent
View All Shipments
Generate Reports

Module 6: Billing

Auto Invoice Generation
Amount + Tax Calc
PDF Download
Email Invoice

COURIER MANAGEMENT - FINAL DOCUMENTATION

Conclusion & Future Scope (Page 3 of 3)

Results & Achievements

- 100% Modules Completed
- 10/10 Test Cases Passed
- 5 Tables, 8 APIs Working
- Booking Flow: <30 sec
- Tracking Real-time <1 sec
- Ready for Deployment

Challenges Faced & Fixed

- Tracking ID Duplicate -> UNIQUE constraint
- Cost Calc 0kg Error -> Validation weight>0
- OTP Not Expiring -> 5min timer added
- Mobile UI Breaking -> Bootstrap responsive

Future Scope

- Mobile App Android/iOS
- AI Route Optimization
- Real GPS Live Tracking
- Payment Gateway
- Multi-language Tamil/English

Project Deliverables

- Source Code: GitHub Repo
- PPT: 15 Slides Presentation
- Report: 40 Pages PDF
- Demo Video: 5 Minutes
- DB: MySQL Dump File
- Certificate: SmartBridge